COL761: Assignment 2
K-means clustering in high-dimensions

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(a) Reasoning for the Selected Value of k

Based on the empirical evidence generated by our K-means implementation, the optimal number of clusters selected for both Dataset 1 (shape: 362×9) and Dataset 2 (shape: 387×12) is $k = 5$.

This determination was made using the **Elbow Method**. The algorithm computes the k-means objective function, which minimizes the within-cluster sum of squared Euclidean distances:

$$\min_{C} \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$$

In our implementation, this objective value is extracted using the `kmeans.inertia_` attribute from the `scikit-learn` library, which directly calculates this exact formulation.

As observed in Figure 1, which is plotted on a logarithmic scale to handle the high variance in early k values, the inertia drops drastically for $k \in \{1, 2, 3, 4\}$. The rate of decrease sharply attenuates at $k = 5$, forming an "elbow." The automated heuristic in the script (analyzing the second discrete difference of the inertia curve via `np.diff`) corroborated this visual inflection point, mathematically isolating $k = 5$ as the point where adding more clusters yields diminishing returns in variance reduction.

(b) Structural and Statistical Assumptions Underlying K-means

K-means implicitly relies on strong geometric and statistical assumptions about the underlying data distribution:

- **Spherical (Isotropic) Clusters:** Because it uses unweighted Euclidean distance, K-means assumes that the clusters are spherical and that variance is symmetric across all dimensions. It essentially models the data as a mixture of Gaussian distributions with identity covariance matrices ($\Sigma = \sigma^2 I$).
- **Similar Cluster Sizes and Densities:** The algorithm assumes that all clusters contain roughly the same number of data points and possess similar densities. It heavily penalizes large variances, meaning dense, tight clusters can disproportionately draw centroids away from sparse, spread-out clusters.
- **Hard Assignments:** It assumes every data point belongs explicitly and equally to exactly one cluster, with no probabilistic "soft" overlap.

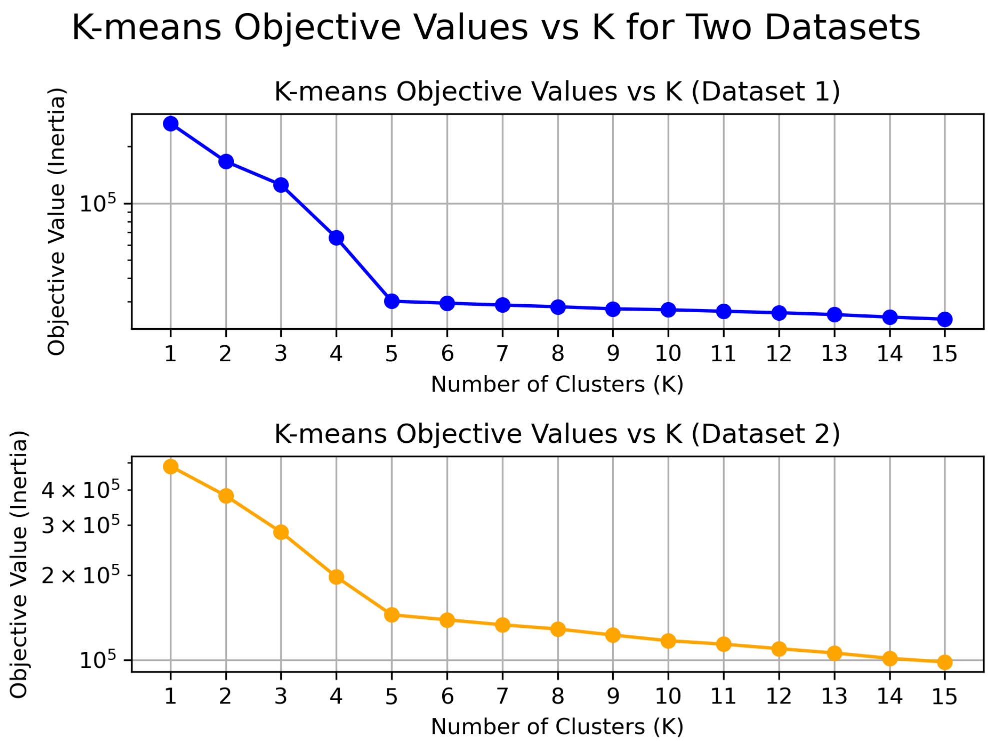

Figure 1: K-means Objective Values (Inertia) vs K for Dataset 1 and Dataset 2.

(c) Limitations of the Model Selection Procedure

Relying solely on the visual or heuristic-based Elbow Method has notable limitations, particularly in high dimensions ($d > 3$):

- **Subjectivity:** The "elbow" is often ambiguous. In high-dimensional spaces, the curse of dimensionality can cause the distances between points to converge, making the inertia curve relatively smooth without a sharp, defining kink.
- **Lack of Penalty for Complexity:** The inertia function is monotonically decreasing. It will inevitably reach 0 when $k = n$. The elbow method lacks a formal penalization term for model complexity, unlike criteria such as AIC or BIC.
- **Brittle Heuristics:** The automated second-derivative proxy used in the script is highly sensitive to minor local fluctuations in the objective curve and can fail if the inertia drops smoothly rather than sharply.

(d) Methodological Improvements and Alternative Validation Principles

To robustly validate the choice of k, the following alternative methodologies should be considered:

- **Silhouette Analysis:** Silhouette Score as the primary metric evaluates both intra-cluster cohesion and inter-cluster separation. A higher score strictly indicates better-defined clusters.

- **The Gap Statistic:** This formalizes the elbow method by comparing the total within-cluster variation for different values of k with their expected values under a null reference distribution of the data (i.e., a uniform distribution with no obvious clustering). The optimal k is where this gap is maximized.
- **Davies-Bouldin Index:** An internal evaluation scheme where a lower score indicates a model with better separation between the generated clusters.

(e) Failure Modes of K-means and Detection Procedures

When K-means is inappropriate: K-means fails catastrophically when its structural assumptions are violated. Well-known failure modes include non-convex cluster shapes (e.g., concentric circles or interlocking "moons"), clusters with heavily skewed or anisotropic covariances (elongated ellipses), and datasets with massive disparities in cluster density.

References

[1] Scikit-learn documentation: *sklearn.cluster.KMeans.*
Retrieved from: `https://scikit-learn.org/stable/modules/generated/sklearn.cluster.KMeans.html`

[2] GeeksforGeeks: *Elbow Method for optimal value of k in KMeans.*
Retrieved from: `https://www.geeksforgeeks.org/machine-learning/elbow-method-for-optimal-value-of-k-in-kmeans/`

[3] Hastie, T., Tibshirani, R., & Friedman, J. (2009). *The Elements of Statistical Learning.* Springer.